

1. Top-Level Architectural Concept & Value Proposition

TRACEID solves identity verification without LLM hallucination or fake confidence percentages. The LLM assists with extraction and report prose, but **case status decisions are 100% deterministic** (written exclusively in `core/status.py`).

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Zero Identity Fabrication

System returns `INSUFFICIENT EVIDENCE` when data is sparse. It refuses to guess or invent identity links.

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D9 Source Independence

MinHash shingling groups news mirrors & web copies into clusters. 3 copies of 1 bio count as 1 confirmation, not 3.

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Consent & Security Perimeter

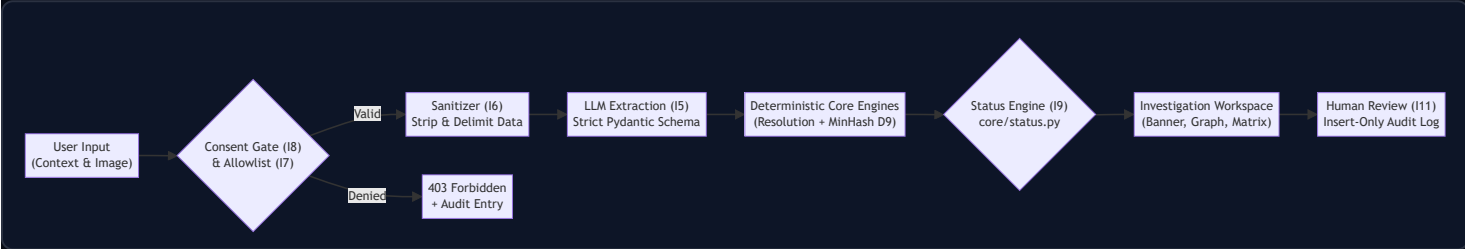
Mandatory consent gate (I8), domain allowlist (I7), and sanitizer wrapping payload in data block (I6 injection defense).

⚖️

Human-in-the-Loop Audit

Reviewers confirm/reject claims with mandatory reasons. Actions are insert-only; evidence is never deleted (I11).

2. High-Level Data & Processing Flowchart



3. The 5 Locked Case Statures (Decision Rules v1)

Status	Deterministic Rule Criteria	Jury Takeaway
STRONG MATCH	≥2 Independent Clusters + ≥1 Discriminating Signal + Margin ≥1 over runner-up. Zero hard contradiction.	High-confidence identity proof verified across separate sources.
POSSIBLE MATCH	≥1 Independent Cluster + Discriminating or multiple Corroborating signals.	Supported identity match, recommended for secondary review.
AMBIGUOUS	≥2 Candidates reach POSSIBLE match with margin < 1 (No separating signal).	System detects name collision and refuses to guess. Shows runner-up.
INSUFFICIENT EVIDENCE	No evidence, only weak signals (name/username similarity), or 0 discriminating clusters.	Sparse digital footprint. "Unknown" is a 1st-class valid output.
LIKELY DIFFERENT	Unresolved hard contradiction found (e.g. impossible concurrent physical timeline).	Factual contradiction blocks match automatically.

4. 11-Stage Pipeline State Machine



5. The 5 Golden Jury Scenarios (1-Click Replay in UI)

Scenario	Subject / Context	Status Result	What It Proves to the Jury
Scenario A	Dr. Aarav Mehta (NEURAX Keynote)	STRONG MATCH	Multi-cluster academic profile. Web mirrors collapsed into 1 cluster via shingling.
Scenario B	Priya Nair (AI Safety Symposium)	AMBIGUOUS	Two distinct professionals (Bengaluru vs Pune). System returns BOTH candidates side-by-side.
Scenario C	Marcus Vance (Freelance)	INSUFFICIENT EVIDENCE	Sparse footprint. Explicit breakdown of what was checked vs missing. Zero hallucination.
Scenario D	Viktor Brand (CyberCorp Payload)	INSUFFICIENT EVIDENCE	Adversarial prompt injection intercepted by sanitizer (I6 active defense). Attacker payload ignored.
Scenario E	Sarah Jenkins (Bar Association)	LIKELY DIFFERENT	Hard contradiction (full-time Tokyo civil service vs London High Court magistrate 2022-2025).

6. Alignment with Jury Evaluation Criteria

💡 Technical Innovation & Architecture

Combines FastAPI async pipeline, MinHash/Jaccard shingling for source independence, and React Flow graph visualizer.

🔒 Security & Privacy First

Consent gate (I8), domain allowlists (I7), EXIF stripping, and prompt injection defense (I6). All actions audit-logged.

🎯 Accuracy & Interpretability

No arbitrary percentage dials. Clear matrix of Discriminating, Corroborating, and Weak evidence signals.

🧠 Human Control & Oversight

Insert-only human review actions (Confirm/Reject). Status recomputed instantly without destroying raw evidence.

the 1990s, the number of people in the world who are undernourished has increased from 600 million to 800 million (FAO 1996).

There are a number of reasons why the world's population is becoming more undernourished. The most important is the rapid increase in the world's population. The world's population is now over 6 billion and is projected to reach 9 billion by the year 2050 (UN 1994).

Another reason is the increasing demand for food. As the world's population grows, the demand for food increases. This is especially true in the developing countries, where the population is growing rapidly.

A third reason is the increasing demand for meat. As the world's population grows, the demand for meat increases. This is especially true in the developed countries, where the population is growing slowly.

A fourth reason is the increasing demand for fish. As the world's population grows, the demand for fish increases. This is especially true in the developing countries, where the population is growing rapidly.

A fifth reason is the increasing demand for fruit and vegetables. As the world's population grows, the demand for fruit and vegetables increases. This is especially true in the developed countries, where the population is growing slowly.

A sixth reason is the increasing demand for grains. As the world's population grows, the demand for grains increases. This is especially true in the developing countries, where the population is growing rapidly.

A seventh reason is the increasing demand for oil. As the world's population grows, the demand for oil increases. This is especially true in the developed countries, where the population is growing slowly.

A eighth reason is the increasing demand for minerals. As the world's population grows, the demand for minerals increases. This is especially true in the developing countries, where the population is growing rapidly.

A ninth reason is the increasing demand for energy. As the world's population grows, the demand for energy increases. This is especially true in the developed countries, where the population is growing slowly.

A tenth reason is the increasing demand for water. As the world's population grows, the demand for water increases. This is especially true in the developing countries, where the population is growing rapidly.

A eleventh reason is the increasing demand for land. As the world's population grows, the demand for land increases. This is especially true in the developing countries, where the population is growing rapidly.

A twelfth reason is the increasing demand for forests. As the world's population grows, the demand for forests increases. This is especially true in the developed countries, where the population is growing slowly.

A thirteenth reason is the increasing demand for oceans. As the world's population grows, the demand for oceans increases. This is especially true in the developing countries, where the population is growing rapidly.

A fourteenth reason is the increasing demand for mountains. As the world's population grows, the demand for mountains increases. This is especially true in the developed countries, where the population is growing slowly.

A fifteenth reason is the increasing demand for rivers. As the world's population grows, the demand for rivers increases. This is especially true in the developing countries, where the population is growing rapidly.

A sixteenth reason is the increasing demand for lakes. As the world's population grows, the demand for lakes increases. This is especially true in the developed countries, where the population is growing slowly.

A seventeenth reason is the increasing demand for wetlands. As the world's population grows, the demand for wetlands increases. This is especially true in the developing countries, where the population is growing rapidly.

A eighteenth reason is the increasing demand for deserts. As the world's population grows, the demand for deserts increases. This is especially true in the developed countries, where the population is growing slowly.

A nineteenth reason is the increasing demand for tundra. As the world's population grows, the demand for tundra increases. This is especially true in the developing countries, where the population is growing rapidly.

A twentieth reason is the increasing demand for glaciers. As the world's population grows, the demand for glaciers increases. This is especially true in the developed countries, where the population is growing slowly.

A twenty-first reason is the increasing demand for icebergs. As the world's population grows, the demand for icebergs increases. This is especially true in the developing countries, where the population is growing rapidly.

A twenty-second reason is the increasing demand for snow. As the world's population grows, the demand for snow increases. This is especially true in the developed countries, where the population is growing slowly.

A twenty-third reason is the increasing demand for rain. As the world's population grows, the demand for rain increases. This is especially true in the developing countries, where the population is growing rapidly.

A twenty-fourth reason is the increasing demand for clouds. As the world's population grows, the demand for clouds increases. This is especially true in the developed countries, where the population is growing slowly.

A twenty-fifth reason is the increasing demand for wind. As the world's population grows, the demand for wind increases. This is especially true in the developing countries, where the population is growing rapidly.

A twenty-sixth reason is the increasing demand for sun. As the world's population grows, the demand for sun increases. This is especially true in the developed countries, where the population is growing slowly.